

Capstone Project: End-to-End Sales Analysis

Financial Sample dataset (700 transactions, Sep 2013 - Dec 2014)

1. Executive Summary (for stakeholders)

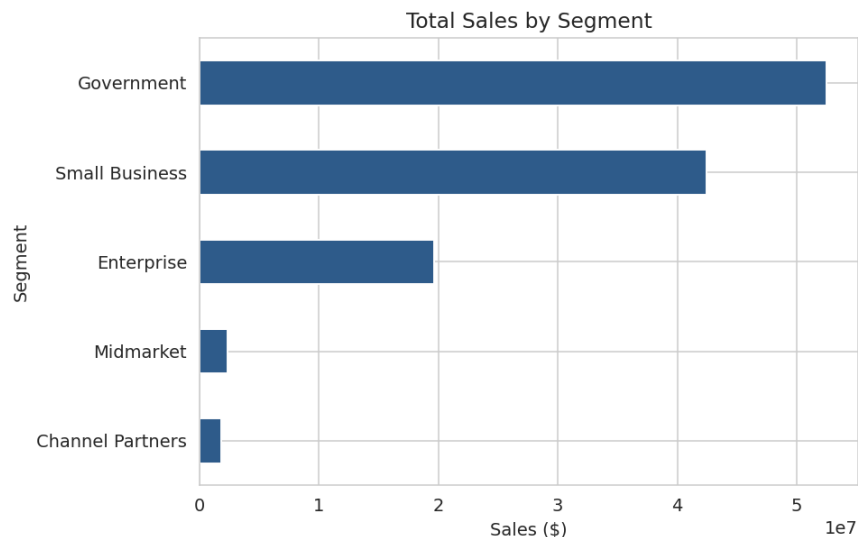
The business generated **\$118.7M in sales** and **\$16.9M profit** (14.2% margin). Government is the anchor segment (44% of sales, 22% margin). Enterprise is the main concern: \$19.6M in sales but a **net loss of \$614K**. Channel Partners is small but earns a ~73% margin. Heavy discounting (High band, ~12.5% average rate) cuts profit, and sales peak strongly in October and December. A Random Forest model predicts sales with $R^2 = 0.98$.

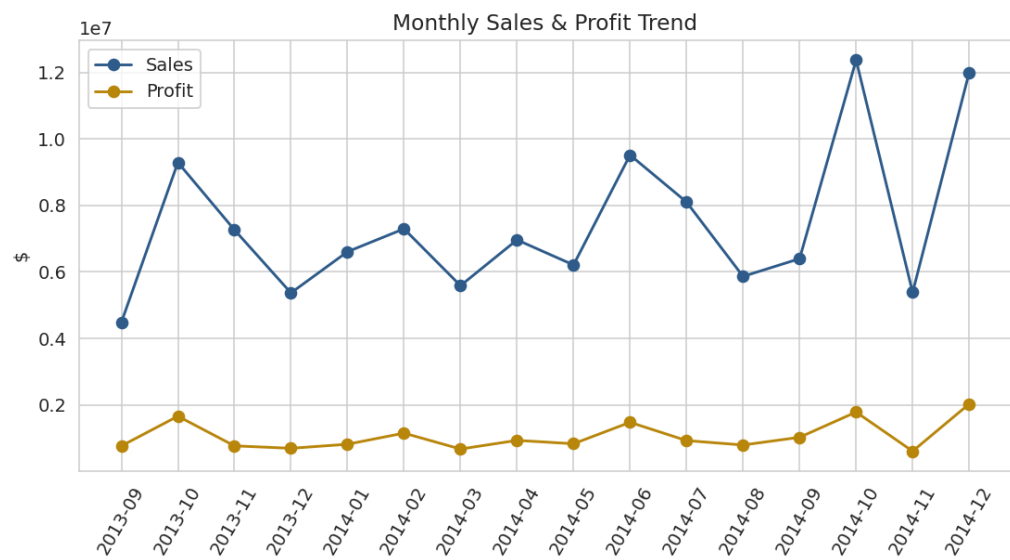
2. Approach

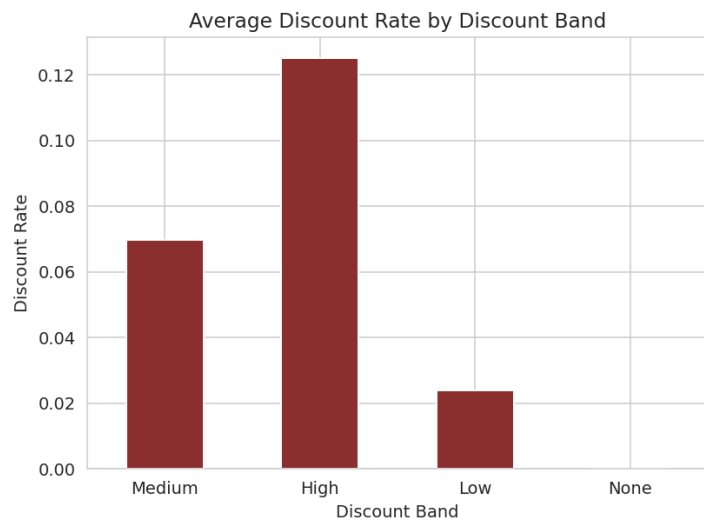
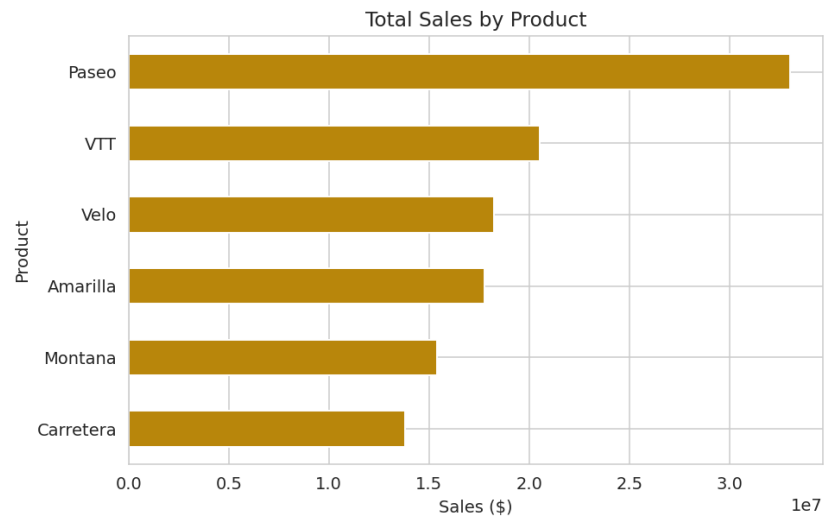
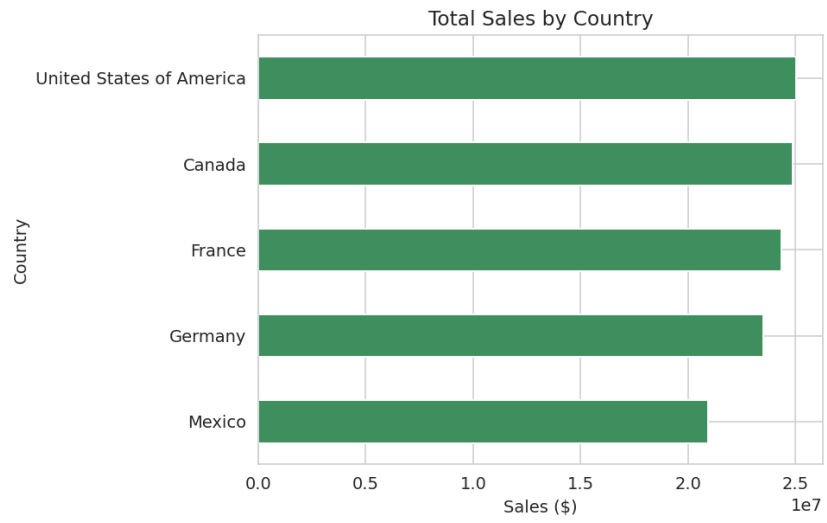
Clean: 53 blank Discount Band values were set to "None" (no discount); dates parsed; discount rate and profit margin added. **Analyze:** Pandas and SQL (SQLite) group-by analysis. **Visualize:** Excel charts, HTML dashboard, and Power BI (steps in README). **Model:** regression to predict Sales.

3. Exploratory Data Analysis

Segment	Sales	Profit	Margin
Government	\$52.5M	\$11.4M	21.7%
Small Business	\$42.4M	\$4.1M	9.8%
Enterprise	\$19.6M	-\$0.6M	-3.1%
Midmarket	\$2.4M	\$0.7M	27.7%
Channel Partners	\$1.8M	\$1.3M	73.1%



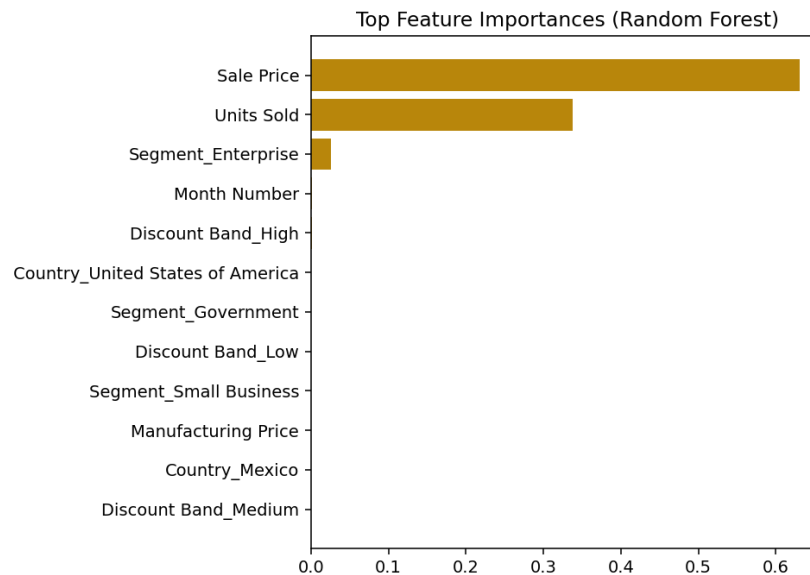
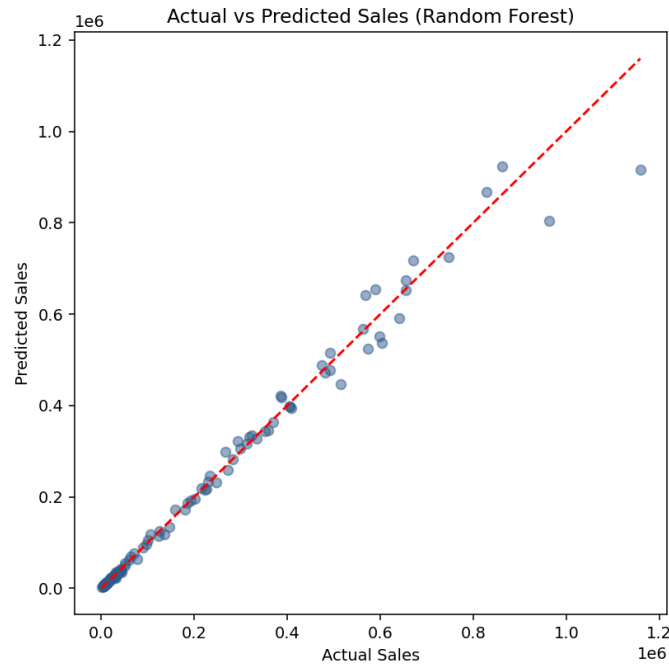




4. Regression Model: Predicting Sales

Target: Sales. Predictors: Segment, Country, Product, Discount Band, Units Sold, Sale Price, Manufacturing Price, Month. Gross Sales, Discounts and COGS were excluded because they are derived from Sales (data leakage). 80/20 train/test split.

Model	R2	MAE	RMSE
Linear Regression	0.762	\$82,989	\$114,708
Random Forest (best)	0.984	\$11,154	\$30,031



5. Findings and Recommendations

Finding	Recommendation
Government = 44% of sales, 22% margin	Protect and grow: prioritize renewals and dedicated coverage.
Enterprise loses \$614K on \$19.6M sales	Review pricing and discounts; renegotiate or cap discounts.
Channel Partners margin ~73%, tiny volume	Invest to scale this efficient channel.
High discount band ~12.5% rate erodes profit	Require approval above the Medium band.
Peaks in Oct and Dec	Plan inventory, staffing and marketing around seasonality.
Paseo top product; Amarilla best margin (15.9%); Carretera lowest sales, 13.2% margin	Prioritize Paseo and Amarilla; review Carretera pricing and costs.
Model R2 = 0.98	Use it to forecast revenue from planned units and prices.

6. Limitations

Only 16 months of data; the model was tested on a random split, not future periods. Margins here are from transaction-level data and may differ from company accounts.